Imperial College London
AERO96004 Aircraft Structures – Plate Bending & Buckling

Tutorial 1 – Plate Bending

Questions

A rectangular simply-supported plate has length a in the x-direction and width b in the y-direction. Assume that the plate has a known flexural stiffness D. The plate is subject to a concentrated force F (i.e. a point load) acting at the location $x=x_0$, $y=y_0$.

1) Consider the general Navier's series expansion of the applied pressure load:

$$p(x,y)=\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}B_{jk}\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}$$

Find the coefficients B_{mn} in the expression above corresponding to the applied point load F. *(see the hint on page 2 of this handout for how to proceed)*

[30 marks]

2) Find the analytical expression of the transverse displacement $w(x,y)$ for the simply supported plate described above subject to the force F.

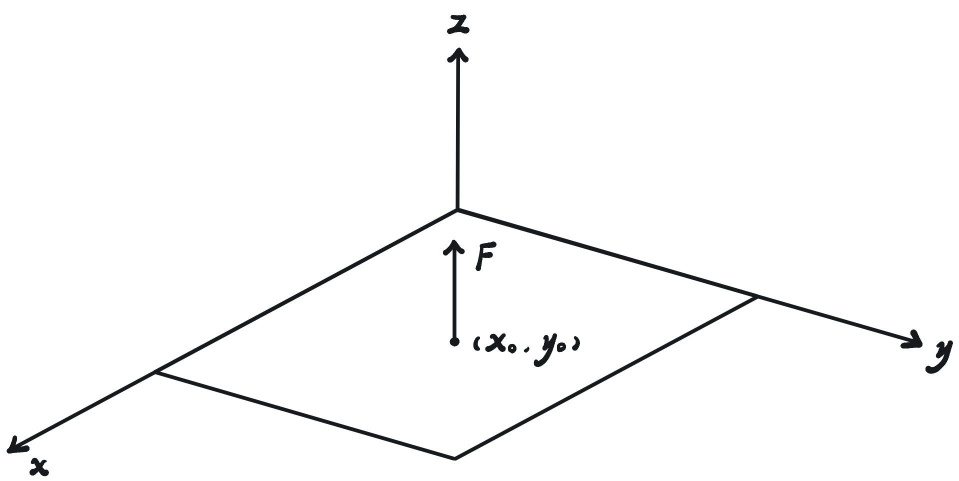
[30 marks]

3) Determine the maximum deflection for a square plate ($b=a$) caused by a point load F applied at its centre ($x_0=y_0=a/2$).

[20 marks]

4) Find the expression of the moments at the centre of the aforementioned square plate.

[20 marks]



1) $p(x,y) = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk} \sin(\frac{j\pi x}{a}) \sin(\frac{k\pi y}{b})$

$\therefore \int_0^a\int_0^b p(x,y) \sin(\frac{j'\pi x}{a}) \sin(\frac{k'\pi y}{b})\, dx\, dy$

$= \int_0^a\int_0^b \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk} \sin(\frac{j\pi x}{a}) \sin(\frac{j'\pi x}{a}) \sin(\frac{k\pi y}{b}) \sin(\frac{k'\pi y}{b})\, dx\, dy$

$= B_{jk} (\frac{a}{2})(\frac{b}{2})$

$\therefore B_{jk} = \frac{4}{ab}\int_0^a\int_0^b p(x,y) \sin(\frac{j\pi x}{a}) \sin(\frac{k\pi y}{b})\, dx\, dy$

$= \frac{4}{ab}\int_0^a\int_0^b F\delta(x-x_0)\delta(y-y_0) \sin(\frac{j\pi x}{a}) \sin(\frac{k\pi y}{b})\, dx\, dy$

$= \frac{4F}{ab} \sin(\frac{j\pi x_0}{a}) \sin(\frac{k\pi y_0}{b})$

2) Let $w = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} A_{jk} \sin(\frac{j\pi x}{a}) \sin(\frac{k\pi y}{b})$

for z-direction force equilibrium,

$-\frac{Et^3}{12(1-\nu^2)}(\frac{\partial^4 w}{\partial x^4} + 2\frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4}) + p = 0$

$\therefore \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} A_{jk}\pi^4(\frac{j^4}{a^4} + 2\frac{j^2}{a^2}\frac{k^2}{b^2} + \frac{k^4}{b^4}) \sin(\frac{j\pi x}{a}) \sin(\frac{k\pi y}{b}) = \frac{12p(1-\nu^2)}{Et^3}$

$\therefore \int_0^a\int_0^b LHS \sin(\frac{j'\pi x}{a}) \sin(\frac{k'\pi y}{b})\, dx\, dy = \int_0^a\int_0^b RHS \sin(\frac{j'\pi x}{a}) \sin(\frac{k'\pi x}{b})\, dx\, dy$

According to the answer 1),

$(\frac{a}{2})(\frac{b}{2}) A_{jk}\pi^4(\frac{j^4}{a^4} + 2\frac{j^2}{a^2}\frac{k^2}{b^2} + \frac{k^4}{b^4}) = \frac{1}{D}(\frac{a}{2})(\frac{b}{2}) B_{jk}$

$\therefore A_{jk} = \frac{4F\sin(\frac{j\pi x_0}{a})\cos(\frac{k\pi y_0}{b})}{abD\pi^4(\frac{j^2}{a^2} + \frac{k^2}{b^2})^2}$

$\therefore w = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4F}{\pi^4 D ab[(\frac{j}{a})^2 + (\frac{k}{b})^2]^2} \sin(\frac{j\pi x}{a}) \sin(\frac{j\pi x_0}{a}) \sin(\frac{k\pi y}{b}) \sin(\frac{k\pi y_0}{b})$

3) for $a=b$, $x_0 = y_0 = \frac{a}{2}$

$w = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4F}{\pi^4 D a^2[(\frac{j}{a})^2 + (\frac{k}{a})^2]^2} \sin(\frac{j\pi x}{a}) \sin(\frac{j\pi}{2}) \sin(\frac{k\pi y}{a}) \sin(\frac{k\pi}{2})$

$\therefore \frac{\partial w}{\partial x} = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4F}{\pi^4 D a^2[(\frac{j}{a})^2 + (\frac{k}{a})^2]^2}(\frac{j\pi}{a}) \cos(\frac{j\pi x}{a}) \sin(\frac{j\pi}{2}) \sin(\frac{k\pi y}{a}) \sin(\frac{k\pi}{2}) = 0$

$\therefore x|_{w_{max}} = \frac{a}{2}$

similarily, $y|_{w_{max}} = \frac{b}{2}$

$$\therefore w(\frac{a}{2}, \frac{b}{2}) = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4F}{\pi^4 D a^2 [(\frac{j}{a})^2 + (\frac{k}{a})^2]^2} \sin^2(\frac{j\pi}{2}) \sin^2(\frac{k\pi}{2})$$

$$= \begin{cases} 0 & , j \text{ or } k \text{ is even} \\ \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4F}{\pi^4 D (\frac{j^2}{a} + \frac{k^2}{a})^2} & , j \text{ and } k \text{ are odd} \end{cases}$$

$$\therefore w_{max} = \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2}$$

4) $M_x = -D(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2})$

$M_y = -D(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2})$

$M_{xy} = -D(1-\nu)\frac{\partial^2 w}{\partial x \partial y}$

where

$$\frac{\partial w}{\partial y} = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2} (\frac{k\pi}{a}) \sin(\frac{j\pi x}{a}) \sin(\frac{j\pi}{2}) \cos(\frac{k\pi y}{a}) \sin(\frac{k\pi}{2})$$

$$\therefore \frac{\partial^2 w}{\partial x^2} = (\frac{j\pi}{a})^2 w$$

$$\frac{\partial^2 w}{\partial y^2} = (\frac{k\pi}{a})^2 w$$

$$\frac{\partial^2 w}{\partial x \partial y} = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2} (\frac{jk\pi^2}{a^2}) \cos(\frac{j\pi x}{a}) \sin(\frac{j\pi}{2}) \cos(\frac{k\pi y}{a}) \sin(\frac{k\pi}{2})$$

$$\therefore \frac{\partial^2 w}{\partial x^2}\Big|_{(\frac{a}{2}, \frac{b}{2})} = (\frac{j\pi}{a})^2 \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2}$$

$$\frac{\partial^2 w}{\partial y^2}\Big|_{(\frac{a}{2}, \frac{b}{2})} = (\frac{k\pi}{b})^2 \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2}$$

$$\frac{\partial^2 w}{\partial x \partial y}\Big|_{(\frac{a}{2}, \frac{b}{2})} = 0$$

$$\therefore M_x = -D[(\frac{j\pi}{a})^2 \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2} + \nu (\frac{k\pi}{b})^2 \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2}]$$

$$= -D \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4F(j^2+\nu k^2)}{\pi^2 D (j^2+k^2)^2}$$

$$M_y = -D[(\frac{k\pi}{b})^2 \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2} + \nu (\frac{j\pi}{a})^2 \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4Fa^2}{\pi^4 D (j^2+k^2)^2}]$$

$$= -D \sum_{j=1,3,5\ldots}^{\infty} \sum_{k=1,3,5\ldots}^{\infty} \frac{4F(k^2+\nu j^2)}{\pi^2 D (j^2+k^2)^2}$$

$$M_{xy} = 0$$

Hint: The load due to a concentrated *F* force acting on the plate surface at the point *(x_0, y_0)* can be expressed via the two-dimensional Dirac's delta function as:

$$p(x,y) = F\delta(x-x_0)\delta(y-y_0)$$

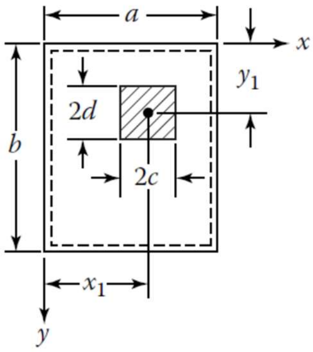
Remember that the Dirac's delta function satisfies the properties:

P1) $$\int_0^a\int_0^b \delta(x-x_0)\delta(y-y_0)\,dx\,dy = 1;$$

P2) $$\int_0^a\int_0^b f(x,y)\delta(x-x_0)\delta(y-y_0)\,dx\,dy = f(x_0,y_0),$$

where $f(x,y)$ is an arbitrary continuous function defined in the domain $[0,a]\times[0,b]$. Hence, we have this useful result:

$$\int_0^a\int_0^b \delta(x-x_0)\delta(y-y_0)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\,dx\,dy = \sin\frac{m\pi x_0}{a}\sin\frac{n\pi y_0}{b}.$$



$$p(x,y) = \sum\sum p_{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$F = \int_A p(x,y)dxdy \Rightarrow let p(x,y) = \frac{F}{4cd} \Rightarrow$$

$$F = F = \int_{x_1-c}^{x_1+c}\int_{y_1-d}^{y_1+d}\frac{F}{4cd}dxdy = F\int_{x_1-c}^{x_1+c}\int_{y_1-d}^{y_1+d}\frac{1}{4cd}dxdy$$

$$let\ c,d = 0\ but\ keep\ the\ area\ under\ integral\ constant\ as = 1 \Rightarrow$$

$$point\ load \Rightarrow F = \int_{x_1}^{x_1}\int_{y_1}^{y_1} F\delta(x-x_1)\delta(y-y_1)dxdy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F\delta(x-x_1)\delta(y-y_1)dxdy$$

$$p(x,y) = F\delta(x-x_1)\delta(y-y_1)$$

OR, Alternative Solutuion

$$p_{mn} = \frac{P}{abcd}\int_{y_1-d}^{y_1+d}\int_{x_1-c}^{x_1+c}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\,dx\,dy$$

$$p_{mn} = \frac{4P}{\pi^2 mncd}\sin\frac{m\pi x_1}{a}\sin\frac{n\pi y_1}{b}\sin\frac{m\pi c}{a}\sin\frac{n\pi d}{b}.$$

Tutorial 1, Plate Bending

Preliminary Considerations (Revisiting L)

First of all, let us introduce the Kronecker delta, which is defined as follows

$$\delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases}$$

where m and n are positive integers. Let us also recall the following orthogonality property of sine functions

$$\int_0^a \sin\frac{m\pi x}{a} \sin\frac{n\pi x}{a} dx = \frac{a}{2}\delta_{mn}$$

Following Navier's approach, the pressure acting on a simply-supported plate is expanded as a double sine series in x and y, i.e.

$$p(x,y) = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk} \sin\frac{j\pi x}{a} \sin\frac{k\pi y}{b}$$

In order to find the coefficients B_{jk} we can multiply both terms of the last equation by sine harmonics in x and y; thus

$$p(x,y)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b} = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk} \sin\frac{j\pi x}{a} \sin\frac{k\pi y}{b}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

Integrating the above with respect to the plate area we have

$$\int_0^a\int_0^b p(x,y)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}dx\,dy =$$
$$= \int_0^a\int_0^b \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk} \sin\frac{j\pi x}{a} \sin\frac{k\pi y}{b}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}dx\,dy$$

Hence

$$\int_0^a\int_0^b p(x,y)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}dx\,dy =$$
$$= \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk} \int_0^a \sin\frac{j\pi x}{a}\sin\frac{m\pi x}{a}dx\int_0^b \sin\frac{k\pi y}{b}\sin\frac{n\pi y}{b}dy$$

Exploiting the orthogonality property of the sine harmonics we obtain

$$\int_0^a\int_0^b p(x,y)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}dx\,dy = \frac{ab}{4}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty} B_{jk}\delta_{jm}\delta_{jk} = \frac{ab}{4}B_{mn}$$

Hence, we have the following fundamental result

$$B_{mn} = \frac{4}{ab}\int_0^a\int_0^b p(x,y)\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}dx\,dy$$

Similarly, in Navier's approach the transverse displacement of a simply supported plate is expanded as a double sine series

$$w(x,y)=\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}w_{jk}\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}$$

Note that the series above satisfies exactly the boundary conditions of the problem for a simply-supported plate, i.e. zero displacements and zero bending moments (zero curvatures) on all edges.

The exact relation between the w_{jk} and B_{jk} coefficients is found by substituting the sine series into the plate equilibrium equation

$$\frac{\partial^4 w}{\partial x^4}+2\frac{\partial^4 x}{\partial x^2\,\partial y^2}+\frac{\partial^4 w}{\partial y^4}=\frac{p}{D}$$

Substituting the displacement expression into the left hand side of the last equation, we find

$$\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}w_{jk}\left[\left(\frac{j\pi}{a}\right)^2+\left(\frac{k\pi}{b}\right)^2\right]^2\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}=\frac{p}{D}$$

Multiplying both terms by sine harmonics in x and y and integrating with respect to the plate area, we obtain

$$\int_0^a\int_0^b\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}w_{jk}\left[\left(\frac{j\pi}{a}\right)^2+\left(\frac{k\pi}{b}\right)^2\right]^2\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}dxdy=$$
$$=\frac{1}{D}\int_0^a\int_0^b\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}p\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

Exploiting the orthogonality property of sine functions, we have

$$\frac{ab}{4}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}w_{jk}\left[\left(\frac{j\pi}{a}\right)^2+\left(\frac{k\pi}{b}\right)^2\right]^2\delta_{jk}\delta_{jm}=\frac{1}{D}\int_0^a\int_0^b\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}p\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

Therefore

$$w_{mn}\left[\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]^2=\frac{1}{D}\frac{4}{ab}\int_0^a\int_0^b\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}p\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

Note that on the right hand side we now have the B_{mn} coefficients divided by the flexural stiffness of the plate, hence

$$w_{mn}=\frac{1}{D}\left[\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]^{-2}B_{mn}$$

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Tutorial 1, Plate bending

Solutions

Q1) As outlined in the tutorial handout, a concentrated force F acting on the plate can be expressed as an "impulse" in 2D space as

$$p(x,y) = F\delta(x - x_0)\delta(y - y_0)$$

where $\delta(x-x_0)$ and $\delta(y-y_0)$ are the Dirac delta functions with respect to x and y. The B_{jk} coefficients are given by

$$B_{jk} = \frac{4}{ab}\int_0^a \int_0^b p(x,y)\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}dx\,dy$$

Substituting the pressure associated with a concentrated force into the last equation we have

$$B_{jk} = \frac{4F}{ab}\int_0^a \int_0^b \delta(x - x_0)\delta(y - y_0)\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}dx\,dy$$

Exploiting the properties of the Dirac delta function we obtain

$$B_{jk} = \frac{4F}{ab}\sin\frac{j\pi x_0}{a}\sin\frac{k\pi y_0}{b}$$

Q2) The Navier's expressions of the displacement field for a simply supported plate is

$$w(x,y) = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} w_{jk}\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}$$

where the coefficients of the series are expressed as

$$w_{jk} = \frac{1}{D}\left[\left(\frac{j\pi}{a}\right)^2 + \left(\frac{k\pi}{b}\right)^2\right]^{-2} B_{jk}$$

Substituting the expression of the B_{jk} coefficients into the last equation leads to

$$w_{jk} = \frac{4F}{abD}\left[\left(\frac{j\pi}{a}\right)^2 + \left(\frac{k\pi}{b}\right)^2\right]^{-2}\sin\frac{j\pi x_0}{a}\sin\frac{k\pi y_0}{b}$$

Hence the displacement field is given by

$$w(x,y)=\frac{4F}{abD}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\left[\left(\frac{j\pi}{a}\right)^2+\left(\frac{k\pi}{b}\right)^2\right]^{-2}\sin\frac{j\pi x_0}{a}\sin\frac{k\pi y_0}{b}\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{b}$$

Q3) For a square plate with a point load applied at its centre, we have

$$b=a,\ x_0=y_0=\frac{a}{2}\Rightarrow w(x,y)=\frac{4Fa^2}{\pi^4 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{1}{\left(j^2+k^2\right)^2}\sin\frac{j\pi}{2}\sin\frac{k\pi}{2}\sin\frac{j\pi x}{a}\sin\frac{k\pi y}{a}$$

Note than in the series above all the even terms in j and k are zero.
The maximum displacement is achieved at the centre of the plate, where it is straightforward to check that the derivatives of $w(x,y)$ (i.e. the rotations) are zero; hence

$$w\left(\frac{a}{2},\frac{a}{2}\right)=\frac{4Fa^2}{\pi^4 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{1}{\left(j^2+k^2\right)^2}$$

where only the odd terms must to be considered.

Q4) The second derivatives of the displacement for the square plate are

$$\frac{\partial^2 w}{\partial x^2}=-\frac{4F}{\pi^2 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{j^2}{\left(j^2+k^2\right)^2}\sin\frac{j\pi x_0}{a}\sin\frac{j\pi x}{a}\sin\frac{k\pi y_0}{a}\sin\frac{k\pi y}{a}$$

$$\frac{\partial^2 w}{\partial y^2}=-\frac{4F}{\pi^2 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{k^2}{\left(j^2+k^2\right)^2}\sin\frac{j\pi x_0}{a}\sin\frac{j\pi x}{a}\sin\frac{k\pi y_0}{a}\sin\frac{k\pi y}{a}$$

$$\frac{\partial^2 w}{\partial x\,\partial y}=\frac{4F}{\pi^2 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{jk}{\left(j^2+k^2\right)^2}\sin\frac{j\pi x_0}{a}\cos\frac{j\pi x}{a}\sin\frac{k\pi y_0}{a}\cos\frac{k\pi y}{a}$$

Hence, at the centre of the plate we find

$$\frac{\partial^2 w}{\partial x^2}=-\frac{4F}{\pi^2 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{j^2}{\left(j^2+k^2\right)^2}$$

$$\frac{\partial^2 w}{\partial y^2}=-\frac{4F}{\pi^2 D}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{k^2}{\left(j^2+k^2\right)^2}$$

$$\frac{\partial^2 w}{\partial x\,\partial y}=0$$

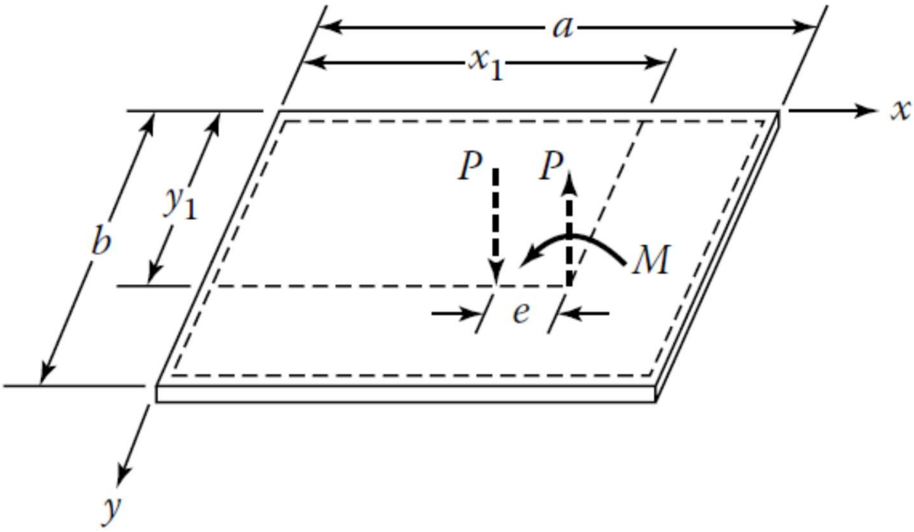
Therefore $M_{xy}(a/2,a/2)=0$. For the bending moments we find

$$M_x=\frac{4F}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{j^2+\nu k^2}{\left(j^2+k^2\right)^2},\qquad M_y=\frac{4F}{\pi^2}\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{\nu j^2+k^2}{\left(j^2+k^2\right)^2}$$

where ν is the Poisson's ratio.

How about this problem:

A simply supported rectangular plate is subjected to a moment M at $x = x_1, y = y_1$
Determine an equation for the deflection.



Let $w = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$

$q = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} B_{mn} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$

for force equilibrium in z - direction

$$-D\left(\frac{\partial^4 w}{\partial x^4} + 2\frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4}\right) + p = 0$$

$$\therefore \int_0^a\int_0^b \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn}\pi^4\left[\frac{m^4}{a^4} + 2\frac{m^2}{a^2}\frac{n^2}{b^2} + \frac{n^4}{b^4}\right] \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b} \sin\frac{m'\pi x}{a} \sin\frac{n'\pi x}{b}\, dx dy$$

$$= \frac{1}{D}\int_0^a\int_0^b \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} B_{mn} \sin\frac{m\pi x}{a} \sin\frac{n\pi x}{b} \sin\frac{m'\pi x}{a} \sin\frac{n'\pi y}{b}\, dx dy$$

for orthogonality.

$$\left(\frac{a}{2}\right)\left(\frac{b}{2}\right) A_{mn}\pi^4\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^2 = \frac{1}{D}\left(\frac{a}{2}\right)\left(\frac{b}{2}\right) B_{mn}$$

$$\therefore A_{mn} = \frac{B_{mn}}{D\pi^4\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]^2}$$

where

$$\int_0^a\int_0^b q \sin\frac{m'\pi x}{a}\sin\frac{n'\pi y}{b}\,dx\,dy$$

$$= \int_0^a\int_0^b \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} B_{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\sin\frac{m'\pi x}{a}\sin\frac{n'\pi y}{b}\,dx\,dy$$

for orthogonality.

$$\int_0^a\int_0^b q \sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\,dx\,dy = \left(\frac{a}{2}\right)\left(\frac{b}{2}\right)B_{mn}$$

$$\therefore B_{mn} = \frac{4}{ab}\int_0^a\int_0^b q \sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\,dx\,dy$$

decouple M as shown in figure above.

$M = Pe$ for $e \longrightarrow 0$

$$\therefore p = P(x_1, y_1) - P(x_1 - e, y_1)$$

$\therefore B_{mn}$

$$= \frac{4}{ab}\int_0^a\int_0^b [P\delta(x-x_1)\delta(y-y_1) - P\delta(x-x_1+e)\delta(y-y_1)]\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\,dx\,dy$$

$$= \frac{4}{ab}\left[P\sin\frac{m\pi x_1}{a}\sin\frac{n\pi y_1}{b} - P\sin\frac{m\pi(x_1-e)}{a}\sin\frac{n\pi y_1}{b}\right]$$

$$= \frac{4M}{ab}\lim_{e\to 0}\left[\frac{1}{e}\left(\sin\frac{m\pi x_1}{a} - \sin\frac{m\pi(x_1-e)}{a}\right)\sin\frac{n\pi y}{b}\right]$$

$$\therefore w = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{B_{mn}}{D\pi^4\left[\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2\right]^2}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{4M}{D\pi^4 ab\left[\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2\right]^2}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\lim_{e\to 0}\left[\frac{1}{e}\left(\sin\frac{m\pi x_1}{a} - \sin\frac{m\pi(x_1-e)}{a}\right)\sin\frac{n\pi y_1}{b}\right]$$

$$= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{4M}{D\pi^4 ab\left[\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2\right]^2}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\frac{d}{dx_1}\left(\sin\frac{m\pi x_1}{a}\sin\frac{n\pi y_1}{b}\right)$$

* $\dfrac{df}{dx} = \dfrac{f(x+h) - f(x)}{h}$